Experimental Evaluation of Robust Feature Selection

10 October 2026

Abstract

We evaluate robust feature selection using a separate validation of subset search and a frozen study of statistical utility. On 16 predeclared confirmation configurations, storing the dual fractions already obtained by full evaluation reduces complete subset solves by 24.2%, while summed median runtime changes by +4.3%. The reference study compares four synthetic alternatives and two public datasets using independent selection, classifier fitting and testing. Global empirical-risk optimality does not yield uniformly better testing power or prediction. All comparisons, time limits and failures are retained; results from earlier search versions remain labeled separately.

1 Experimental design and implementation

The report separates two bodies of evidence. The frozen reference study evaluates an earlier global-cut implementation with five fractional root rounds. Its statistical outcomes describe those recorded supports, not the current search implementation. The new validation study compares ways of choosing the next binary subset; its rule is selected on development cases and then assessed on separate confirmation cases. The robust objective is shared, but the search versions and timing cohorts are not pooled.

The experiment design follows the distinction between computation, two-sample power and support recovery in Wang et al. [1]. Independent prediction and small-sample error evaluation follow robust-testing studies [2, 3, 4]. These references motivate the measurements; the experiments below are adaptations for a common feature budget rather than replays of published tables.

1.1 Data, splits and fixed settings

Synthetic class zero has independent standard-normal coordinates. Class one changes a hidden signal block: a mean shift of 0.9, standard deviation 1.8, equicorrelation 0.65 with unchanged marginals, or a shared random sign times 1.5 added to its coordinates. Remaining coordinates are standard-normal noise. Positions are shuffled, while recovery uses stable underlying feature identities. The dependence alternative requires joint measurements; the mixture has equal means but different shape and dependence. The primary class radii are both 0.25. Each synthetic seed has 20 alternative and 20 null batches, with

Table 1: Reference statistical data. Selection and classifier fitting counts are per class; final counts show their unit. Repeats are training seeds or overlapping real-data splits.

Data	D/d	Selection / fitting per class	Final observations	Repeats
Synthetic	12/3	20/20	1000/class	30
WDBC	30/3	20/20	489 total	20
Ionosphere	34/3	20/20	271 total	20

40 observations per class in a batch. Radius sensitivity uses 10 seeds per mechanism and radius.

In the reference statistical study, selection, downstream fitting and final observations are disjoint. Centering and sample-standard-deviation scaling are fitted on pooled selection data only and frozen thereafter. Radii, budgets, regularization and bandwidth rules are fixed before final evaluation. The real datasets are UCI WDBC and Ionosphere [5, 6]; identifiers and outcome labels are excluded from the feature matrices.

1.2 Compared selectors

All six practical selectors use the same acquisition budget. Global risk minimizes the sum of class-wise worst-case half-hinge losses over class-conditional empirical Wasserstein balls, with standardized

Table 2: Reference optimization settings. Each row is evaluated separately for mean and mixture alternatives.

D	d	Atoms/class	Seeds	Alternatives
8	2	8	10	Mean, mixture
12	3	8	10	Mean, mixture
20	3	10	5	Mean, mixture
40	3	10	5	Mean, mixture
80	3	10	5	Mean, mixture

L^1 transport. Each selected subset is fully refitted. The finite formulation, dual cuts and whole-space test are described in the accompanying theory note; this report concerns their implementation and evidence. Maximum mean discrepancy (MMD) compares within-class kernel similarities with cross-class similarities to measure a distribution difference. The Gaussian radial-basis-function (RBF) kernel is $k(x, y) = \exp[-\|x - y\|_2^2/(2b^2)]$, where b^2 is a fixed squared bandwidth.

Table 3: Budget-matched selectors and fixed implementation choices.

Selector	Selection procedure
Global risk	Global lower-cut search; actual feasible tests give its upper bound. The reference version has five root rounds and an MMD-screening warm start.
Robust greedy	Forward additions, with a full solve of the same robust objective at every candidate. If time expires, retain the last completed stage and fill by MMD screening.
MMD screening	Rank one-coordinate unbiased Gaussian-kernel MMD estimates; coordinate bandwidths use selection-only median squared cross-class distances.
Wang additive MMD	Paper criterion: additive-kernel mean minus 0.1 times estimated variance, with unit-norm sparse weights. Enumerate supports for $D \leq 12$ and solve their weight problems globally; use forward selection and at most ten improving swaps for larger D.
L1 logistic	Absolute-coefficient ranking from an L1 liblinear fit, $C = 1$, at most 2,000 iterations; ties use selection variance and then coordinate index.
Random	Uniform budget-sized support from a reserved, reproducible random stream.

The Wang implementation reproduces the paper's mathematical criterion, not a literal author-script run. The inspected revision rewards a positive variance term, and its testing routine refits bandwidths using test labels. Our variance term is negative and all testing bandwidths are frozen from selection data. Its native signed additive statistic is reported separately from the common joint RBF test. Known-signal and all-feature supports are diagnostics; the former is unavailable in practice and the latter uses a larger budget.

Every selected support receives the same robust classifier and RBF support-vector machine (SVM), both fitted on the independent downstream sample. The robust classifier uses the same radius and half-hinge objective; the SVM uses $C = 1$ and a downstream-only median bandwidth. This holds the evaluation model fixed while comparing selected coordinates.

1.3 Measurements and uncertainty

Two-sample power is the rejection fraction on independent alternative batches; null rejection uses separate batches drawn from the same class-zero distribution. The common statistic T is an unbiased estimate of squared joint RBF MMD. Its pooled-selection median squared-distance bandwidth is frozen, and 199 fixed-size label permutations give

$$p = \frac{1 + \#\{T_{\text{perm}} \geq T_{\text{observed}} - 10^{-14}\}}{200}, \qquad \text{reject if } p \leq 0.05. \tag{1}$$

Methods share the same batches and permutation seeds. This null rejection rate is distinct from a classifier's class-zero error.

For selected set S and true signal set T_*, recovery is measured by

$$\text{recall} = \frac{|S \cap T_*|}{|T_*|}, \qquad \text{FDP} = \frac{|S \setminus T_*|}{\max\{1, |S|\}}, \qquad \text{NDP} = 1 - \text{recall}. \tag{2}$$

For the six budget-matched synthetic selectors, $|S| = |T_*| = 3$, so precision and F1 equal recall and FDP equals NDP. This equivalence does not apply to the all-feature diagnostic. We show recall rather than repeat equivalent outcomes. Pairwise Jaccard support stability is descriptive and remains in the complete data. No true support is claimed for real datasets.

Let $p_0(x)$ be the robust classifier's probability of class zero. Hard decisions use threshold 0.5, with ties assigned to class zero. Its randomized error and the common hard-classification measure are

$$\begin{aligned} \text{BER}_{\text{hard}} &= \tfrac{1}{2}(\text{FPR} + \text{FNR}), \\ \text{BER}_{\text{random}} &= \tfrac{1}{2}\{\mathbb{E}_0[1 - p_0(X)] + \mathbb{E}_1[p_0(Y)]\}. \end{aligned} \tag{3}$$

FPR is the class-zero probability of predicting class one; FNR is the class-one probability of predicting class zero. The SVM uses decision threshold zero, again assigning ties to class zero. The second equals the half-hinge average; these are not separate endpoints. AUC uses class-one scores. Balanced predictive cross-entropy averages the two classwise negative log probabilities, clipped at 10^{-12}; it evaluates probability quality and is not the optimized half-hinge objective. These secondary outcomes and classwise errors remain in the linked per-trial data.

Optimization outcomes are actual lower and upper bounds, their gap, solved fraction, distinct fully evaluated subsets and charged time. Reference runs use one 30-second selection allowance and declare numerical optimality at gap $10^{-6} + 10^{-5}|\text{UB}|$. Failed attempts and same-model retries are charged and remain in denominators. Synthetic intervals are 95% percentile bootstrap intervals from 2,000 resamples of training seeds; testing batches are first averaged within seed. Real results are mean $\pm$ split SD, since splits overlap.

2 Validation of the revised subset search

The computational question is whether choosing and screening subsets more carefully reduces expensive full evaluations. Each selected binary subset is fully solved, gives an actual feasible test and contributes one tangent. No old test is transferred to a new subset.

The original search uses a fixed Gaussian process (GP) predictor to obtain mean risk and model standard deviation from completed evaluations, then ranks up to eight eligible proposals by mean minus standard deviation. The revision retains the same predictor, proposals and refresh interval. It also stores the safe dual fraction already returned by each full evaluation. A candidate is screened using the largest stored tangent or fraction value. The periodic master also retains these fractions. This uses existing certificates and adds no continuous oracle stage.

The implementation repeats four steps: (1) exclude proposals whose stored lower bound exceeds the incumbent minus half the target tolerance; (2) rank the remaining proposals; (3) fully solve the chosen subset and store its test, tangent and fraction; (4) refresh the global master after three completed evaluations or when no proposal survives. The actual test updates the upper bound; the master supplies the global lower bound and stopping test.

Here T is the recorded support-evaluation count (the solver's visited counter). It is separate from the count meeting inner precision: a limited run may record a final evaluation that misses this target. Every recorded anchor in the current validation meets $\varepsilon/4$, where ε is the target absolute risk tolerance. Physical oracle attempts may include refinements; continuous solves count the primal and dual solver calls inside them. Master calls and master time measure the separate integer-search cost. Numerical accuracy is checked against exhaustive support enumeration, not only the reported outer gap.

2.1 Development and the fixed rule

The 26 development cases use two losses, dimensions 6 and 8 with budgets 2, class sizes 4 and 8, and seeds 3, 7 and 11; two additional cases use dimension 12, budget 3 and four observations per class. These cases use a shared deliberately poor tail-coordinate start. Stage 1 changes the prediction score or gives a refreshed master candidate priority. Stage 2 adds data-ranked proposals or chooses a tangent on the same point's dual optimal face; the latter incurs an extra continuous solve. Stage 3 retains fractions and solves a fraction master at every step. Stage 4 keeps those fractions but refreshes only periodically. Changing the score alone did not reduce evaluations. The last two stages reduce evaluations through stronger stored lower bounds, while their integer masters cost more per call. The stage-4 fraction variant

Table 4: All development comparisons, each on the same 26 paired cases. Stages use separate source snapshots and timing batches. A dash means the measurement was not recorded. An unfinished run remains in its denominator and its partial work is retained.

Stage	Search	T	Continuous solves	Masters	Time (s)	Optimal
1	Mean minus SD	475	—	209	5.188	26/26
1	Expected improvement	475	—	206	4.771	26/26
1	Mean minus SD, master priority	477	—	218	4.859	26/26
1	Improvement, master priority	475	—	217	4.870	26/26
1	Tangent master at every step	467	—	467	6.056	26/26
2	Original GP search	475	950	209	5.072	26/26
2	Ranked proposals	480	960	212	4.884	26/26
2	Dual-face tangent	467	1419	206	17.270	26/26
2	Tangent master at every step	467	934	467	5.817	26/26
3	Original GP search	475	950	209	4.987	26/26
3	Fractions at every step	382	764	382	7.722	26/26
3	Tangent master at every step	467	934	467	5.840	26/26
4	Original GP search	475	950	209	5.132	26/26
4	Stored fractions	398	796	168	4.985	25/26
4	Fractions at every step	382	764	382	7.837	26/26
4	Tangent master at every step	467	934	467	5.988	26/26

has one solver-reported infeasible master and 25 of 26 optimal results; its partial total is not an all-success improvement claim. The returned incumbent satisfies the stored lower constraints. The implementation now verifies such a witness and retries the same master without presolve inside the remaining allowance. This numerical repair is applied to every compared method. The rule was fixed before confirmation; the following integration replay checks the repaired implementation on the old development cases.

Table 5: Integration replay of the fixed rule after numerical repair: 26 cases, one run per method. Totals include every run; this is development verification, not a second confirmation cohort.

Search	T	Continuous solves	Masters	Master time (s)	Total time (s)	Optimal runs
Tangent master at every step	467	934	467	3.948	7.558	26/26
Original GP search	475	950	209	2.205	5.822	26/26
Stored fractions	398	796	168	2.944	6.135	26/26

2.2 Independent confirmation

The 16 predeclared confirmation configurations use $(D, d) = (7, 2)$ or $(10, 3)$, six observations per class, seeds 19 and 23, radii $(0.12, 0.20)$ or $(0.30, 0.45)$, and half-hinge or logistic loss. They reuse 4 gener-

ated datasets across losses and radii. Generated coordinates are used directly, without the standardization fitted in the reference statistical study. Class zero is Gaussian; class one shifts its first two coordinates by 1.4 and 0.9. Nuisance scales span 0.3 to 3. Dimension and budget change the support space; unequal radii and the two losses check both asymmetric uncertainty and linear/conic solver costs. This is an optimization-effort study, not a new power study. All methods use the same data-ranked start. The score numerator is the absolute class-mean difference; its denominator is the sum of both class standard deviations and 10^{-12}. Standard deviations use `ddof=0`; ties are resolved by stable index order. No generating signal identity is used. The allowance is 60 s per run, with one solver thread, no fractional root evaluations, absolute risk tolerance $\varepsilon = 2\times10^{-5}$ and zero relative tolerance. Inner and master precision targets are $\varepsilon/4$. The normalized risk predictor has prior mean $1/2$, covariance $\frac{1}{4}\exp[-\|z - z'\|_1/\max(1,d)]$ and ridge $\max\{10^{-8}, (\varepsilon/(4B))^2\}$, where $B = 1$ for half hinge and $B = 2\log 2$ for logistic. These parameters are fixed. Each method is timed 3 times per configuration, rotating execution order. Times below sum the within-case medians; query counts are unchanged across those repetitions. The 16 configurations are paired settings on 4 datasets, so no independent-trial confidence interval is attached to these totals. Stored fractions reduce total evaluations from 598 to 453 (24.2%), with fewer/equal/more

Table 6: Confirmation totals of per-case medians. All physical oracle attempts equal completed evaluations T; each evaluation makes two continuous solves. Optimal-run counts include all timing repetitions.

Search	T	Continuous solves	Masters	Master time (s)	Total time (s)	Optimal runs
Tangent master at every step	601	1202	601	13.507	17.312	48/48
Original GP search	598	1196	314	8.271	12.715	48/48
Stored fractions	453	906	227	9.610	13.262	48/48

evaluations in 15/0/1 cases. Their total median time is 13.262 versus 12.715 s (+4.3%); they are faster in 8/16 cases. Fewer evaluations therefore do not establish an overall speedup. Master time increases because the stored fraction model is more costly, despite fewer master calls. The unchanged GP cannot be credited as the cause of this query reduction. All 144/144 confirmation runs are numerically optimal and pass binary-only, owned-certificate and actual-incumbent checks. The largest feasible-risk excess over the enumeration lower reference is 1.6×10^{-7}. These checks validate the empirical optimization target; they do not assert population robustness or universal runtime improvement.

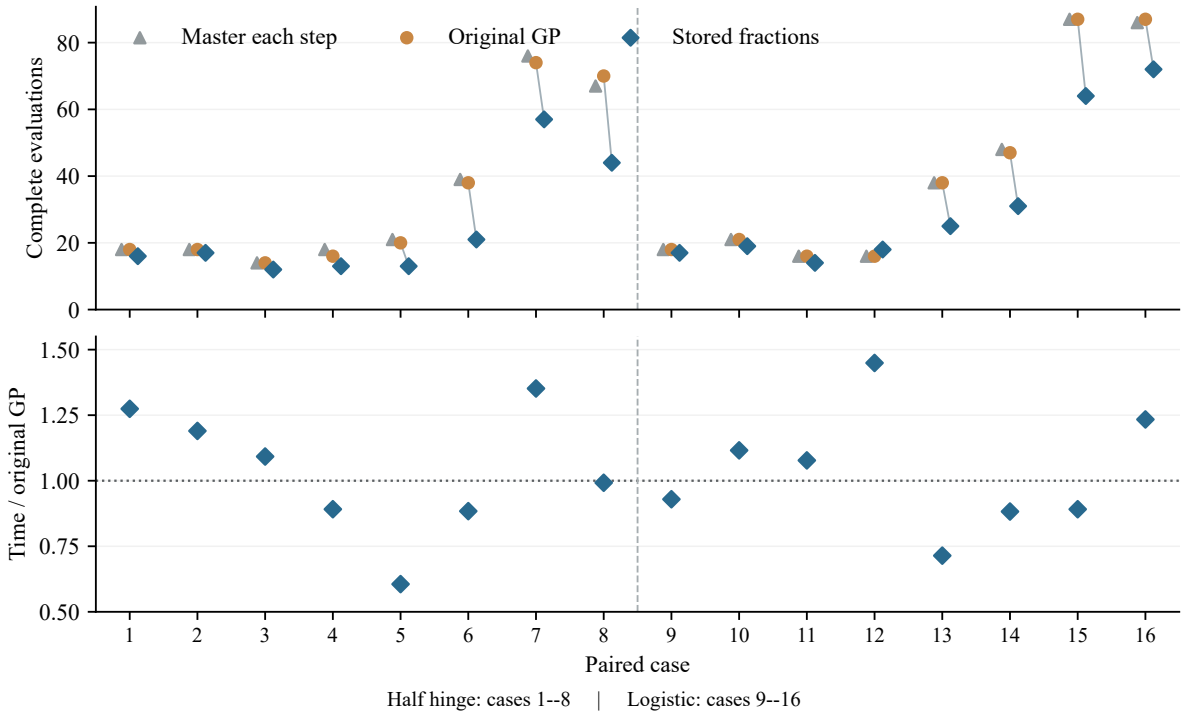


Figure 1: Paired confirmation cases. Upper panel: full binary evaluations (smaller is better). Lower panel: stored-fraction time divided by original GP time, using the three-run median for each method. The dashed line at one means equal time. Cases 1–8 use half hinge and 9–16 logistic; each loss block is ordered by dimension 7/10, seed 19/23, then radius pair.

3 Frozen reference study: computation

The gate comparisons solve the same binary robust objective. Box prices use $w_{kj} \le (B/\theta_k)z_j$, triangle prices use $\theta_0 w_0 + \theta_1 w_1 \le Bz$, and risk-scaled prices replace B by the minimized risk variable. A no-root risk variant isolates the old root-stage effect. All start from the same screened support. Small cases use exhaustive support solutions as a reference; larger cases assess behavior under the common cap. Here $B = 1$ is the constant-test sum-risk bound for half hinge. Across 159 solved gate/enumeration pairs, the largest recorded feasible-objective difference is 0.000, with 0 incompatible intervals at the combined solver tolerances. At $D \ge 20$, global risk reaches the limit in 15/15 mean instances. At $D \ge 20$, global risk reaches the limit in 15/15 mixture instances.

A separately declared complete-fraction comparison uses different seeds and is not pooled with the primary timing cohort. The full fractions are stronger value bounds, but their product-variable integer master can cost more.

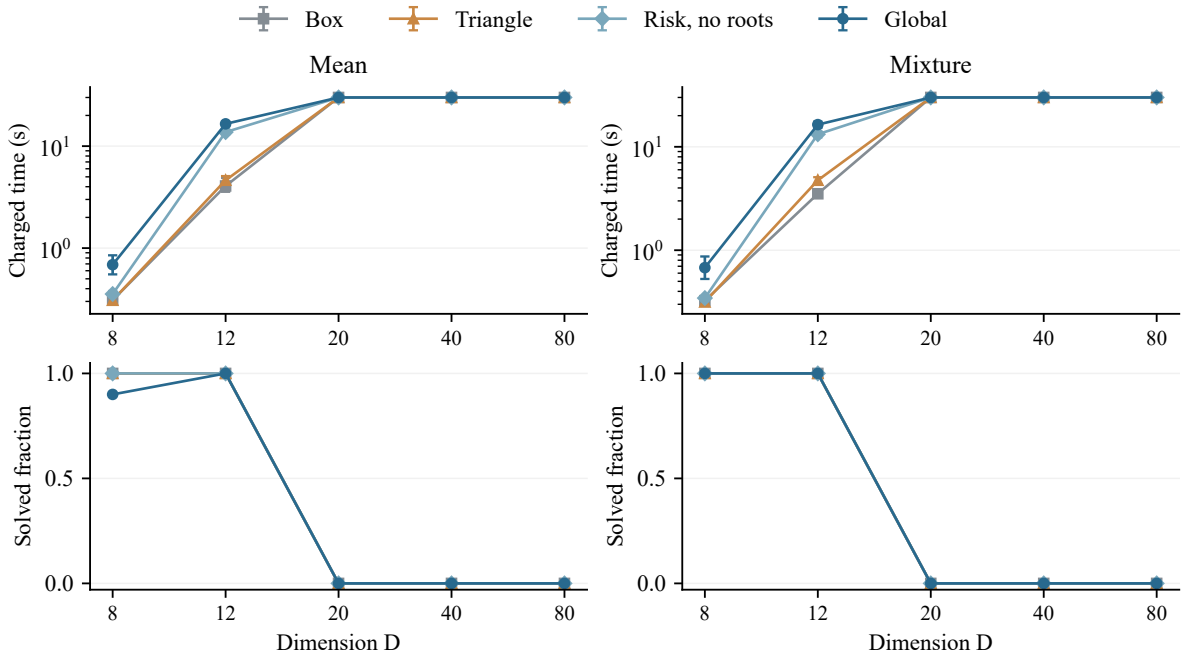


Figure 2: Reference-version charged time and solved fraction. The time axis is logarithmic; intervals resample seeds. Limits and failures remain in solved-fraction denominators; all recorded search, root and retry costs contribute to time.

Table 7: $D = 12$ reference comparison. Time is charged seconds with 95% seed-bootstrap intervals; supports are distinct integer subsets, excluding fractional root calls.

Alternative	Search	Solved	Time (s)	Supports
Mean	Box	10/10	4.07 [3.57, 4.88]	220.0
Mean	Triangle	10/10	4.63 [4.16, 5.08]	219.9
Mean	Risk, no root	10/10	13.73 [13.05, 14.32]	203.2
Mean	Global risk	10/10	16.54 [15.74, 17.28]	197.8
Mixture	Box	10/10	3.52 [3.40, 3.63]	220.0
Mixture	Triangle	10/10	4.75 [4.38, 5.08]	219.2
Mixture	Risk, no root	10/10	13.22 [12.30, 14.07]	200.0
Mixture	Global risk	10/10	16.39 [15.24, 17.52]	197.0

Table 8: Earlier full-fraction supplement. The final column is paired full-fraction minus tangent time (seconds, 95% seed-bootstrap interval); a positive value is slower.

Alternative	D	Tangent solved	Fraction solved	Paired time difference
Mean	8	5/5	5/5	0.19 [0.12, 0.25]
Mean	12	5/5	3/5	9.45 [7.38, 11.39]
Mean	20	0/5	0/5	0.00 [-0.00, 0.00]
Mixture	8	5/5	5/5	0.31 [0.22, 0.40]
Mixture	12	5/5	4/5	10.57 [7.36, 13.79]
Mixture	20	0/5	0/5	0.00 [0.00, 0.00]

4 Frozen reference study: statistical utility

4.1 Synthetic testing and selected signals

For mean data, global power is 0.928, versus 0.992 for marginal screening; the paired difference is -6.3 percentage points [-14.3, -1.3] ($n = 30$). For variance data, global power is 0.360, versus 0.405 for marginal screening; the paired difference is -4.5 percentage points [-19.7, 11.0] ($n = 30$). For dependence data, global power is 0.050, versus 0.053 for marginal screening; the paired difference is -0.3 percentage points [-2.8, 2.0] ($n = 30$). For mixture data, global power is 0.770, versus 0.808 for marginal screening; the paired difference is -3.8 percentage points [-20.3, 11.7] ($n = 30$). All 30/30 mean-shift

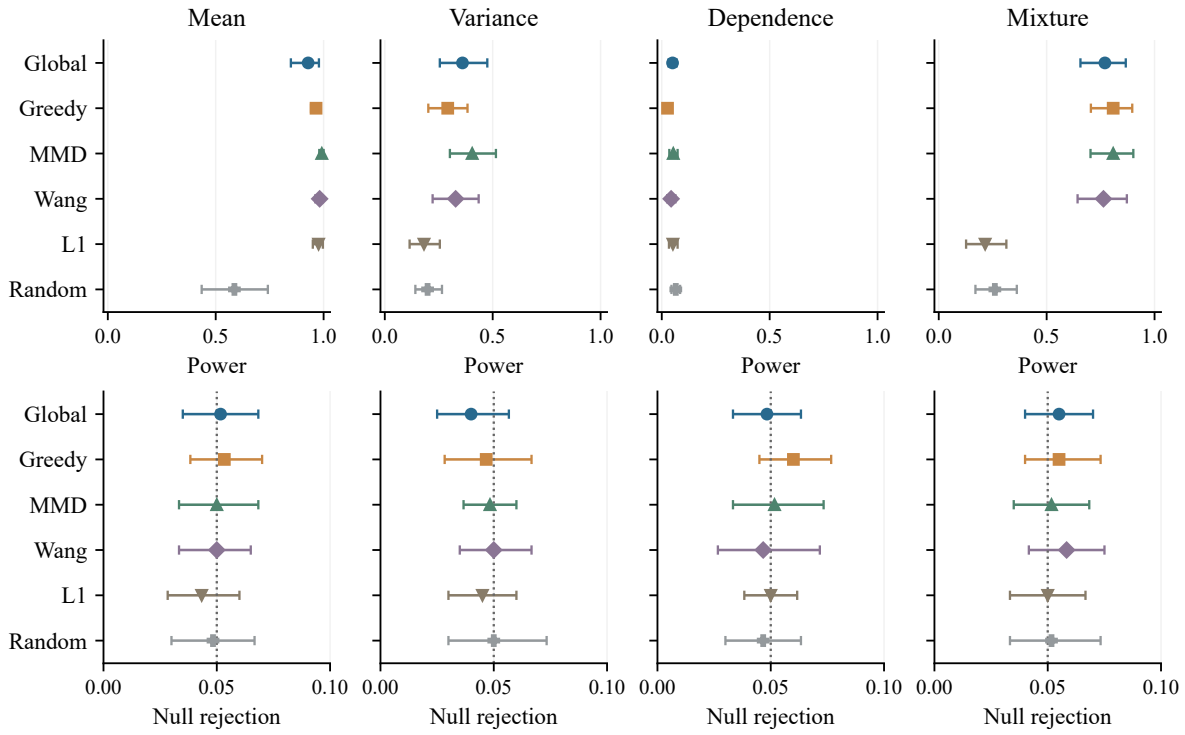


Figure 3: Independent two-sample power (upper row) and null rejection (lower row). The null axis is enlarged around level 0.05, shown by the dotted line; points and intervals use training-seed averages. Only the six budget-matched selectors are compared.

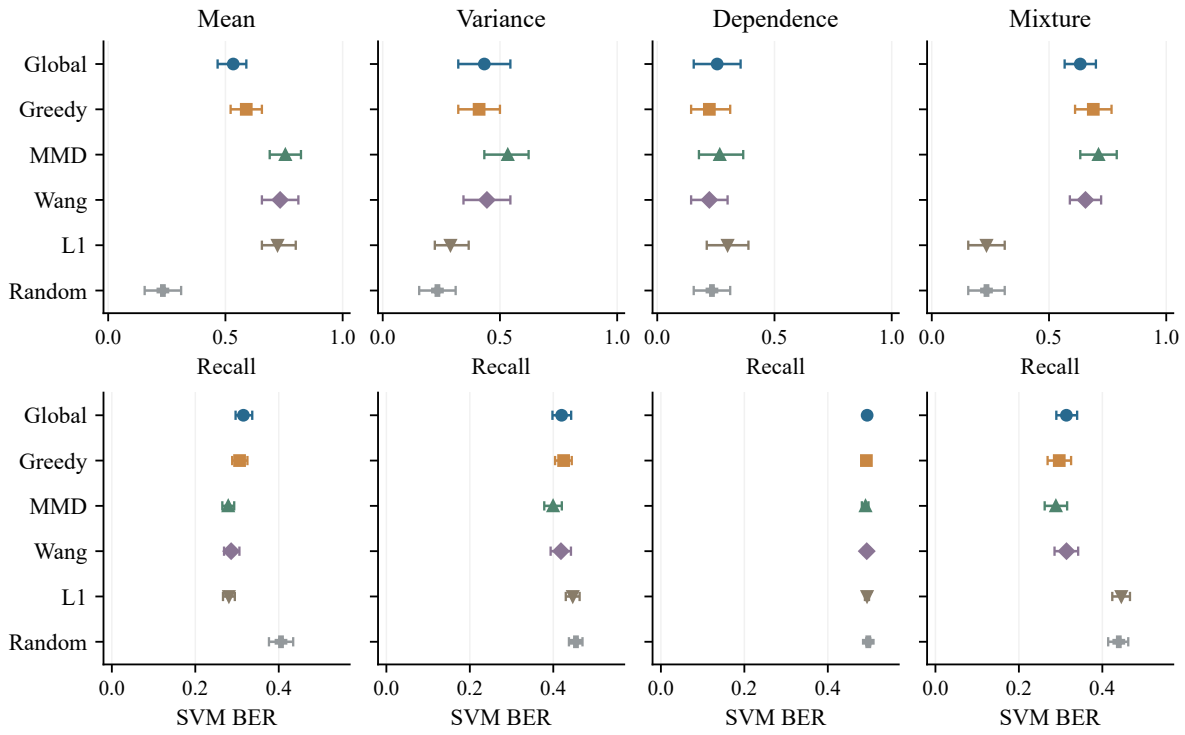


Figure 4: Signal recall (upper row; larger is better) and independent SVM balanced error (lower row; smaller is better). Columns share one mechanism and rows share their metric scale. Intervals resample training seeds.

global searches meet the optimization tolerance, yet greedy power is 0.965 versus 0.928. Empirical-risk optimality does not maximize downstream power. For global selection, 120/120 independent randomized

Table 9: Two evaluators on the Wang-selected supports. Common power uses only the chosen coordinates; native power retains the signed additive weights. Entries are seed means and 95% intervals.

Alternative	n	Common power	Native power	Native null
Mean	30	0.982 [0.962, 0.997]	0.977 [0.950, 0.998]	0.045 [0.025, 0.065]
Variance	30	0.328 [0.222, 0.435]	0.518 [0.408, 0.625]	0.052 [0.033, 0.072]
Dependence	30	0.043 [0.025, 0.063]	0.045 [0.030, 0.060]	0.052 [0.035, 0.070]
Mixture	30	0.763 [0.643, 0.872]	0.945 [0.915, 0.970]	0.040 [0.025, 0.057]

errors exceed their own refitted empirical head bound divided by two. The classifier bound is a different fitted quantity from the selection objective. The population was not established to lie in the fitted ball, so this excess is not a certificate failure.

Table 10: Global-selector refitted head and independent error. Values are paired-seed means; stress is the mean of each seed's maximum over the three preset shifts. Above-bound counts use tolerance 10^{-8}.

Alternative	n	Fit bound / 2	Fresh error	Shift maximum	Above bound
Mean	30	0.251	0.448	0.462	30/30
Variance	30	0.275	0.482	0.484	30/30
Dependence	30	0.287	0.497	0.501	30/30
Mixture	30	0.248	0.423	0.426	30/30

4.2 Radius and measured distribution shifts

Radii 0.10, 0.25 and 0.50 are preset alternatives to the primary radius 0.25. Both the selected support and independently refitted robust classifier change, so this is a comparison of the complete procedure. For population stress, translate each held-out class toward the other along the shared downstream mean-difference direction of L^1 norm one, using magnitudes 0, 0.25 and 0.50. The largest observed error is a maximum over these three shifts, not a Wasserstein supremum. A separate translation of downstream source atoms has an explicit budget-feasible coupling and checks the empirical certificate. Across all

Table 11: Joint radius sensitivity. Entries are seed means and 95% intervals; errors are randomized balanced errors. Stress is the maximum over preset shifts within each seed.

Alternative	θ	n	Power	Fresh error	Shift maximum
Mean	0.10	10	0.790 [0.595, 0.915]	0.494 [0.491, 0.497]	0.496 [0.493, 0.499]
Mean	0.25	10	0.855 [0.650, 0.980]	0.466 [0.450, 0.480]	0.474 [0.461, 0.485]
Mean	0.50	10	0.955 [0.915, 0.990]	0.426 [0.411, 0.440]	0.442 [0.429, 0.453]
Mixture	0.10	10	0.495 [0.295, 0.710]	0.486 [0.478, 0.493]	0.487 [0.480, 0.493]
Mixture	0.25	10	0.630 [0.425, 0.845]	0.446 [0.417, 0.469]	0.448 [0.419, 0.471]
Mixture	0.50	10	0.970 [0.930, 0.995]	0.371 [0.352, 0.392]	0.378 [0.357, 0.401]

1260 budget-feasible source-atom checks, 0 violations exceed 10^{-7} and the largest recorded nonnegative violation is 0.000. This direct check is separate from fresh-population performance.

4.3 Real datasets

Table 12: Independent common classifiers on real datasets. Entries are mean ± overlapping-split SD. Robust error is randomized BER; SVM error is hard BER. These are incumbent supports when search is limited.

Data	Selector	Robust randomized BER	SVM BER	Robust AUC
WDBC	Global risk	0.251±0.046	0.081±0.017	0.965±0.020
WDBC	Robust greedy	0.220±0.032	0.080±0.021	0.973±0.013
WDBC	MMD screening	0.179±0.013	0.084±0.017	0.973±0.011
WDBC	Additive MMD	0.180±0.013	0.082±0.014	0.975±0.010
WDBC	L1 logistic	0.230±0.034	0.078±0.020	0.971±0.017
WDBC	Random	0.310±0.066	0.156±0.078	0.893±0.093
Ionosphere	Global risk	0.300±0.045	0.251±0.094	0.869±0.053
Ionosphere	Robust greedy	0.285±0.035	0.202±0.081	0.885±0.045
Ionosphere	MMD screening	0.284±0.044	0.195±0.053	0.865±0.055
Ionosphere	Additive MMD	0.273±0.042	0.181±0.058	0.877±0.051
Ionosphere	L1 logistic	0.284±0.034	0.198±0.049	0.880±0.046
Ionosphere	Random	0.330±0.028	0.303±0.069	0.835±0.034

Global search is time-limited in 40/40 splits; its supports must not be called certified optima.

5 Recorded outcomes and reproducibility

The 12 original failures were rerun within the unused portion of their original selection allowance: 5 ended optimal, 6 time-limited and 1 failed. The final collection has 3 failures, including later sensitiv-

Table 13: All frozen outcomes retained. Optimum refers to the robust criterion at its declared tolerance. Other includes baseline and diagnostic completion; it does not imply a robust global optimum.

Suite	Records	Optimum	Time limit	Failure	Other
Synthetic	960	114	6	0	840
Optimization	320	199	120	1	0
Real data	240	0	40	0	200
Radius	60	58	0	2	0
Fraction supplement	60	37	23	0	0

ity outcomes. Original and replay costs remain charged. Reference computations used AMD Ryzen 9 7945HX with Radeon Graphics, one solver thread and SciPy-backed HiGHS 1.12.0 (SciPy 1.18.1). Logistic verification uses CVXPY 1.9.3 and Clarabel 0.11.1. Charged selection and end-to-end sums are 2.94 and 2.96 hours, including unsuccessful costs; concurrent-run sums are not elapsed calendar time.

The complete frozen records, all secondary metrics, individual supports, preprocessing and optimization traces remain available in `results/research_20261003/per_trial.csv` and the corresponding JSON records. New development and confirmation evidence has its own frozen directory under `reports/verification/20261010/next_selection`. The report manifest stores input and figure hashes; regeneration validates raw records against configuration hashes and the per-trial table. Neither optimization accuracy nor the fitted empirical-ball certificate establishes population coverage. The observed gaps between the criterion and held-out utility are retained as findings.

References

[1] J. Wang, S. S. Dey, and Y. Xie. *Variable Selection for Kernel Two-Sample Tests*. 2024. arXiv:2302.07415v4. Code inspected at commit `031f740e9ca6` of MMDVar_Selection.

[2] L. Xie, R. Gao, and Y. Xie. *Robust Hypothesis Testing with Wasserstein Uncertainty Sets*. 2021. arXiv:2105.14348.

[3] J. Wang and Y. Xie. *A Data-Driven Approach to Robust Hypothesis Testing Using Sinkhorn Uncertainty Sets*.

[4] F. Zhang, T. Liu, and J. Wang. *Generative Neural Networks for Sinkhorn Distributionally Robust Hypothesis Testing*. 2026. Author code and experiment repository.

[5] UCI Machine Learning Repository. *Breast Cancer Wisconsin (Diagnostic)*. DOI:10.24432/C5DW2B.

[6] UCI Machine Learning Repository. *Ionosphere*. DOI:10.24432/C5W01B.